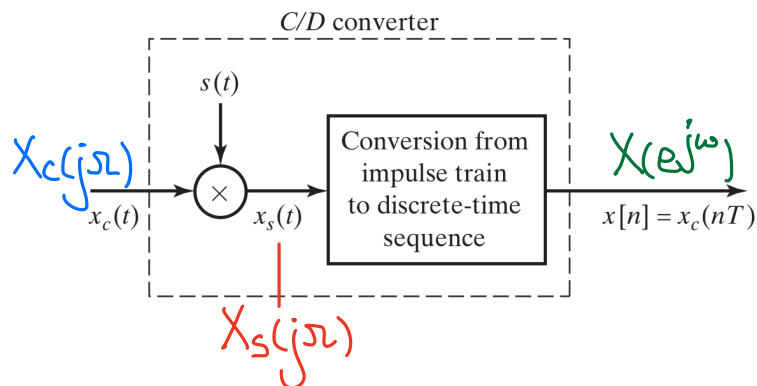


Thursday, September 18

Lecture #8

Homework #4 due 09/25 ; Midterm #1 on 09/30

Last time: sampling continuous-time signals; reconstruction

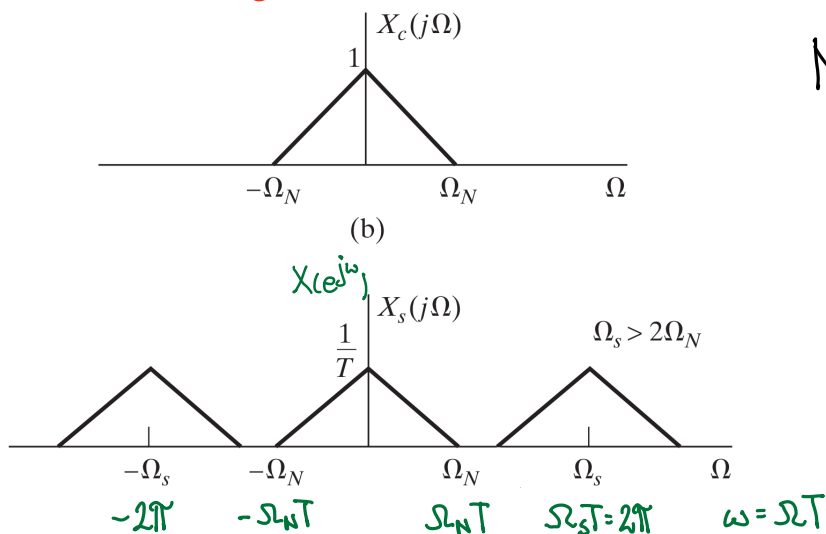


$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

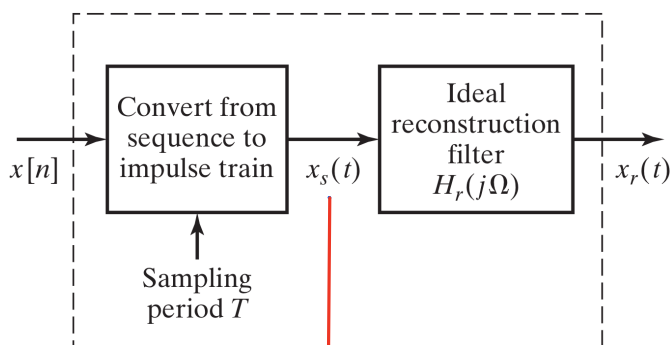
$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\frac{2\pi}{T}))$$

Nyquist theorem:

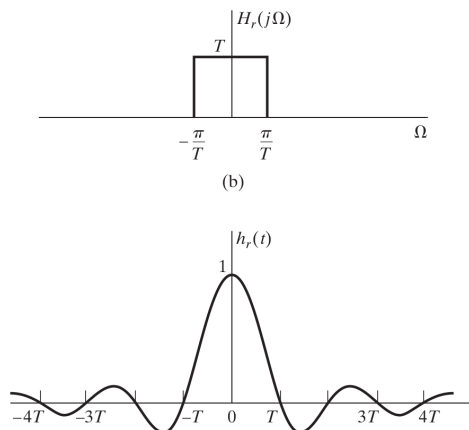
$$\Omega_s = \frac{2\pi}{T} > 2\Omega_N$$



Reconstruction (D/C conversion):



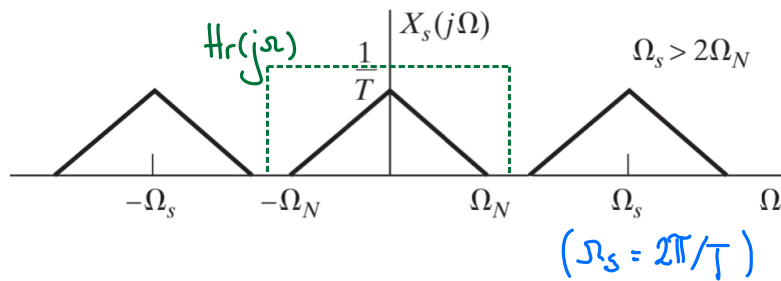
$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT)$$



Why $H_r(j\Omega)$ reconstructs original $x_c(t)$? Recall:

$$X_s(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k\Omega_s))$$

$$\Rightarrow X_r(j\omega) = H_r(j\omega) \cdot \left\{ \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k\Omega_s)) \right\}$$



Only $X_c(j\omega)$ survives filtering of $X_s(j\omega)$ by $H_r(j\omega)$

$$\Rightarrow x_r(t) = x_c(t) \quad (\text{assuming no aliasing})$$

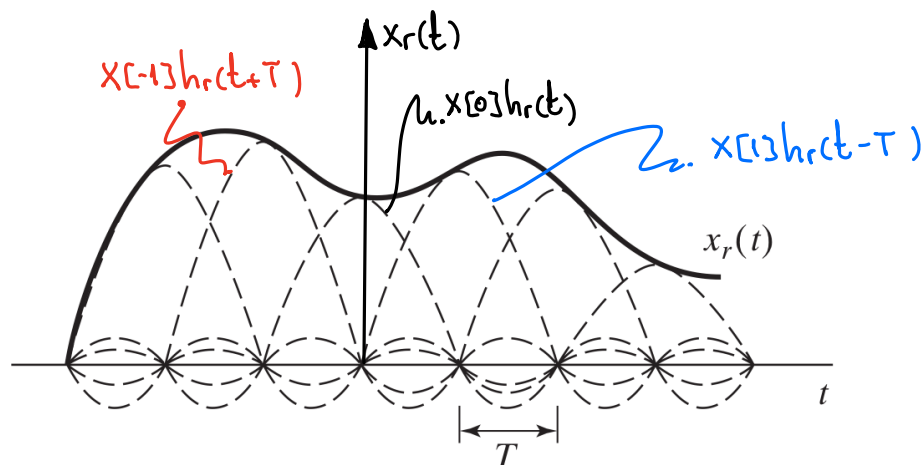
Interpreting signal reconstruction as interpolation:

$$x_r(t) = x_s(t) * h_r(t) = \left(\sum x[n] \delta(t - nT) \right) * h_r(t)$$

$\underbrace{\hspace{10em}}_{= h_r(t - nT)}$

$$\Rightarrow x_r(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t - nT)$$

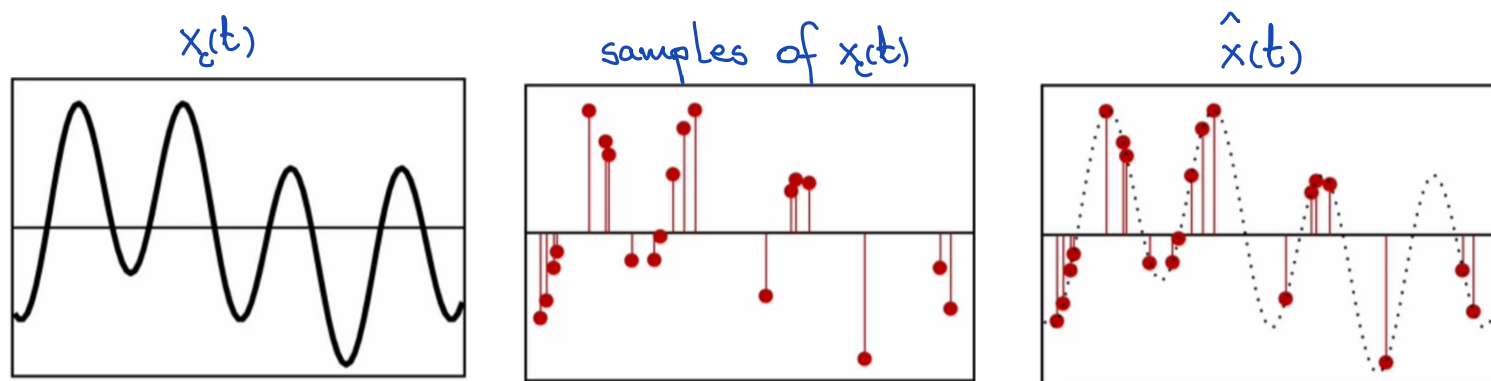
$$= \dots + x[-1] h_r(t+T) + x[0] h_r(t) + x[1] h_r(t-T) + \dots$$



■ Signal reconstruction from non-uniform samples

So far, we assumed uniform sampling (period T)

Sometimes the samples are collected non-uniformly in time:



How do we perform reconstruction now?

Assume $x_c(t)$ has period T ; then

$$x_c(t) = \sum_{k=-M}^M X_k e^{j \frac{2\pi}{T} k t}$$

Here $\{X_{-M}, \dots, X_M\}$ denote Fourier series coefficients

Reconstruction problem formulated as follows:

Given $\{(t_i, x_c(t_i))\}_{i=1}^N$, find $\{\hat{X}_k\}_{k=-M}^M$

The approach: least-squares (i.e., linear regression):

$$\min_{\{X_{-M}, \dots, X_M\}} \sum_{i=1}^N \left| x_c(t_i) - \sum_{k=-M}^M X_k e^{j \frac{2\pi}{T} k t_i} \right|^2 \quad (*)$$

For simplicity, denote

$$y_i = x_c(t_i), \quad K_i = \begin{bmatrix} e^{-j\frac{2\pi}{T}Mt_i} & e^{-j\frac{2\pi}{T}(M-1)t_i} & \dots & e^{j\frac{2\pi}{T}(M-1)t_i} & e^{j\frac{2\pi}{T}Mt_i} \end{bmatrix}^T$$

$$\mathcal{Q} = \begin{bmatrix} X_{-M} & X_{-(M-1)} & \dots & X_{M-1} & X_M \end{bmatrix}^T$$

Then (*) $\Leftrightarrow$

$$\min_{\mathcal{Q}} \sum_{i=1}^N |y_i - \mathcal{Q}^T K_i|^2 \quad (**)$$

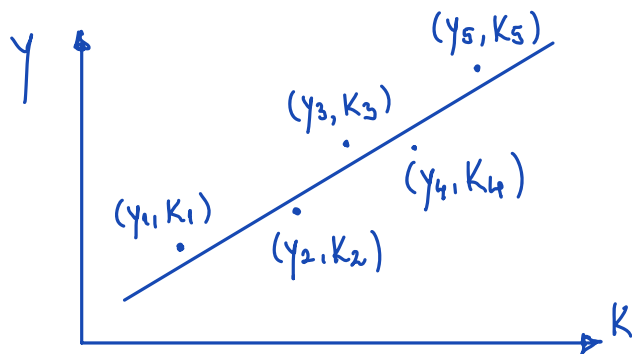
Introduce: $Y = \underbrace{[y_1 \ y_2 \ \dots \ y_N]}_{\text{scalars}}, \quad K = \underbrace{[K_1 \ K_2 \ \dots \ K_N]}_{\text{column-vectors}}$

Then (**) $\Leftrightarrow \min_{\mathcal{Q}} \|Y - \mathcal{Q}^T K\|^2$

$$\Leftrightarrow \min_{\mathcal{Q}} (Y - \mathcal{Q}^T K)^T (Y - \mathcal{Q}^T K)$$

Solving for $\mathcal{Q}$: $\hat{\mathcal{Q}} = (K K^T)^{-1} K Y^T$

An illustration (for scalar $K_i, \mathcal{Q}$):

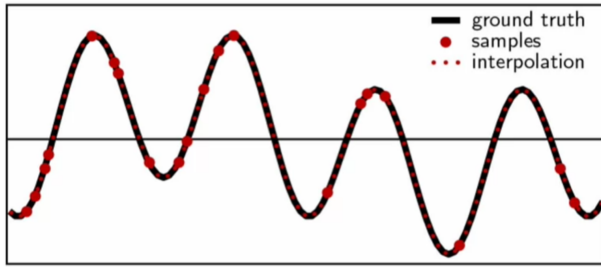


$\mathcal{Q}$: slope of the line

Basically, look for a linear fit to $(y_1, K_1), \dots, (y_N, K_N)$

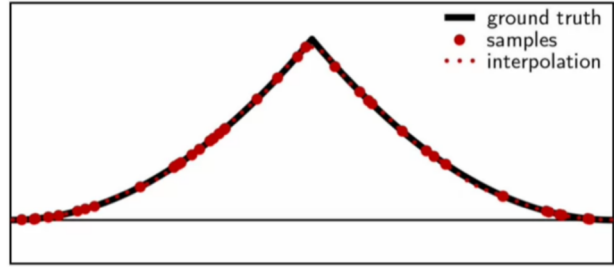
Examples:

$$x(t) = 2\sin(2\pi t) - 4\cos(8\pi t)$$



$$(M=6, N=20)$$

$$x(t) = (1x1 - \frac{1}{2})^2$$



$$(M=10, N=40)$$

Real-world example: interpolating radiation levels from non-uniform measurements following Fukushima accident

2011:

Earthquake, tsunami...

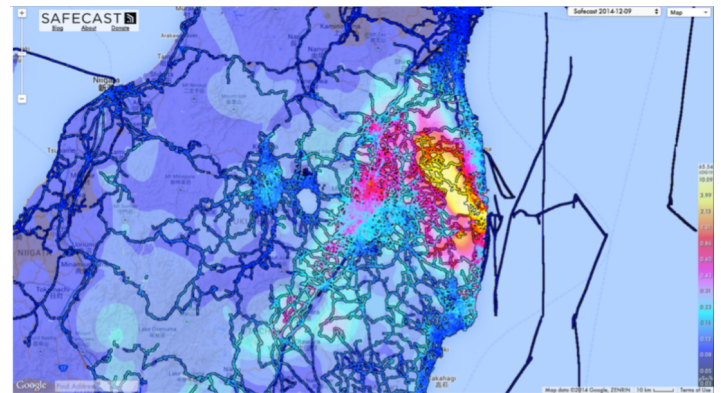
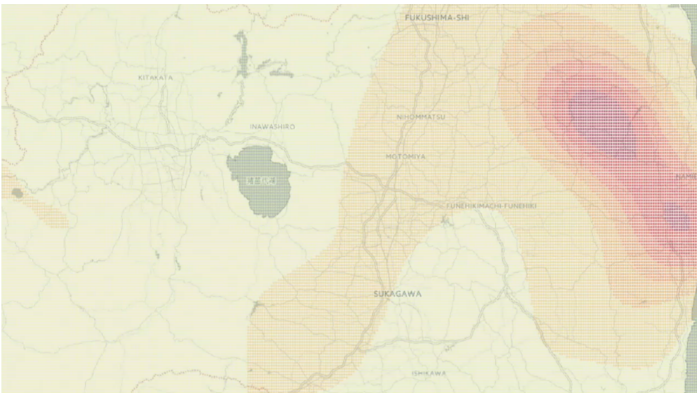


Measurements:



Spatially non-uniform measurements due to being restricted to roads ... challenging interpolation

The result:



$$(M=441, N \approx 10^7)$$

Remark 1: solution to least-squares / Linear regression can be found either in a closed form (with a bit of linear algebra) or numerically (e.g., gradient descent)

Remark 2: the real-world example performs interpolation of a 2-dimensional function and the samples are taken in space rather than in time (i.e., t_i is replaced by $[a_i, b_i]$, where (a_i, b_i) are coordinates where i^{th} measurement is taken).

■ Discrete-time processing of continuous-time signals

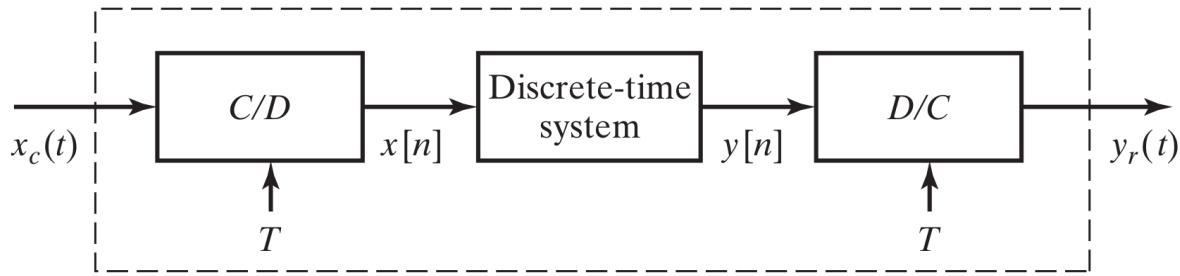
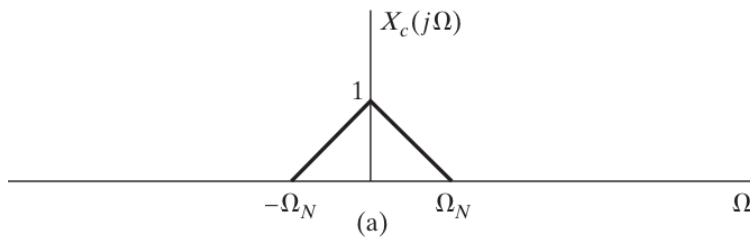
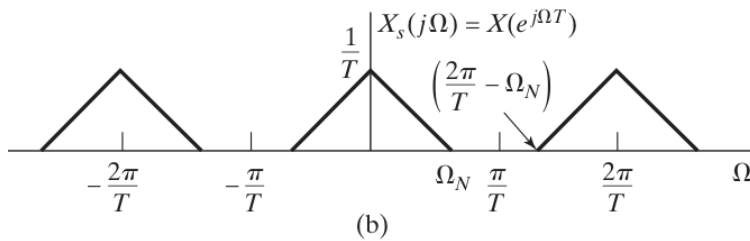


Illustration of low-pass filtering:

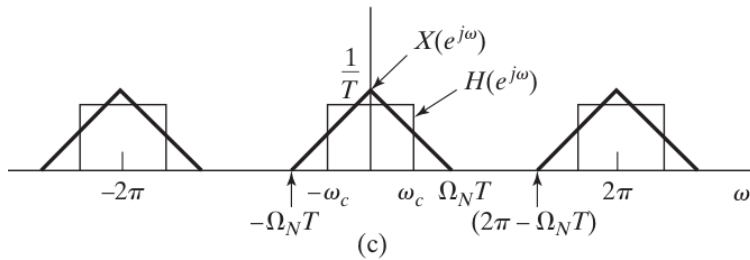


: bandlimited C-T signal



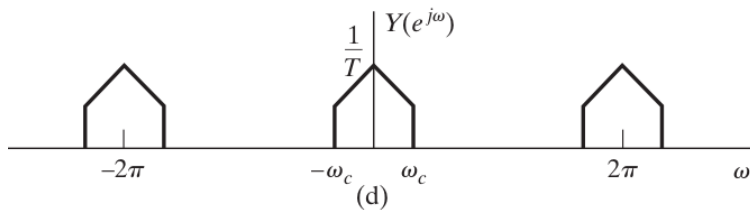
: $X_s(j\Omega) = FT \{x_c(t) s(t)\}$

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

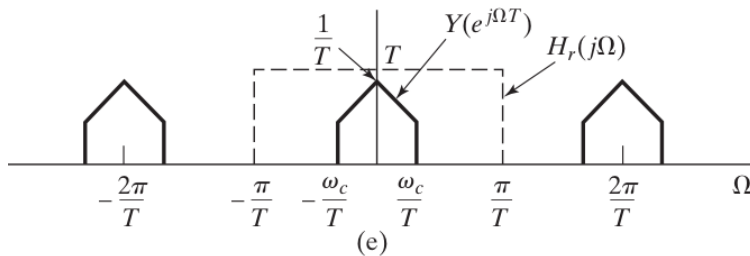


: $X(e^{j\omega}) = DTFT \{x[n]\}$

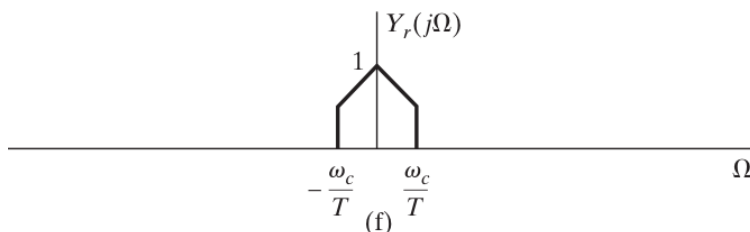
$$\Omega T = \omega$$



: DTFT of $y[n]$



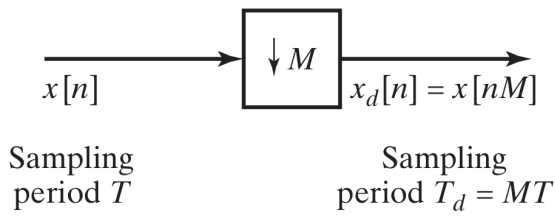
: apply reconstruction filter $H_r(j\Omega)$



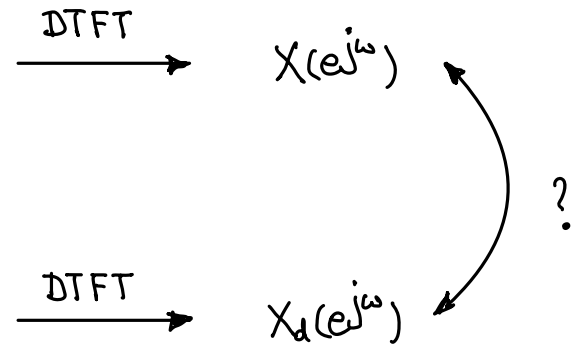
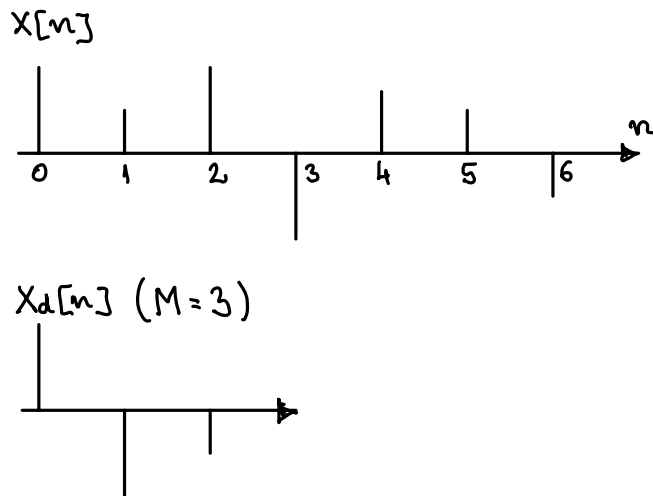
: $FT \{y_r(t)\}$

■ Resampling D-T sequences: downsampling and upsampling

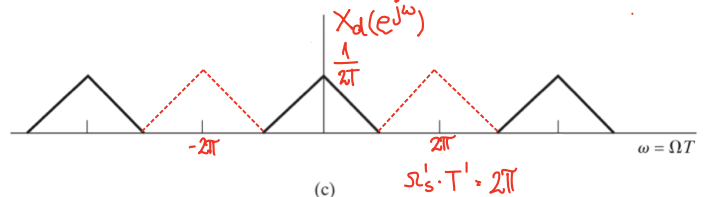
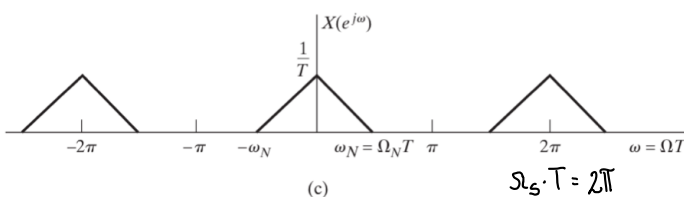
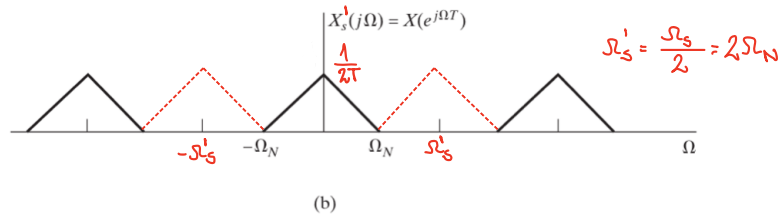
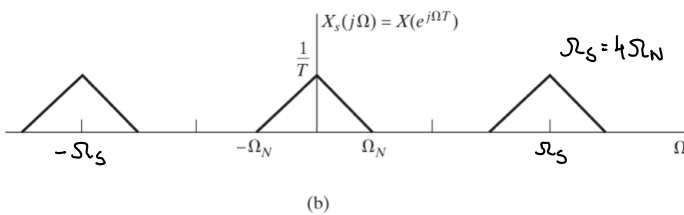
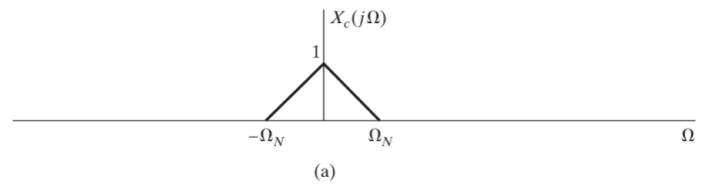
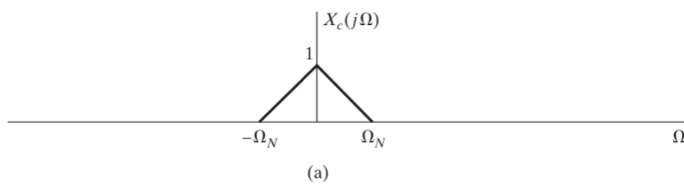
Downsampling: reducing sampling rate by an integer factor



sampling rate
compressor

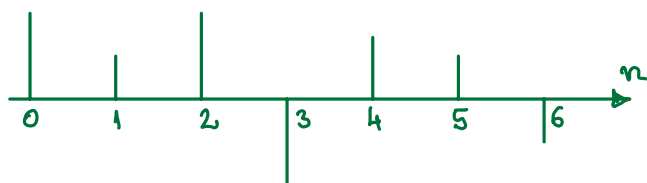


An illustration: assume signal is originally sampled at $\Omega_s = 4\Omega_N$ (no aliasing!), downsample it with $M=2$



Formal derivation of the relationship between $X(e^{j\omega})$ & $X_d(e^{j\omega})$:

$x[n]$



DTFT $\rightarrow$

$X(e^{j\omega})$

$x_d[n] (M=3)$



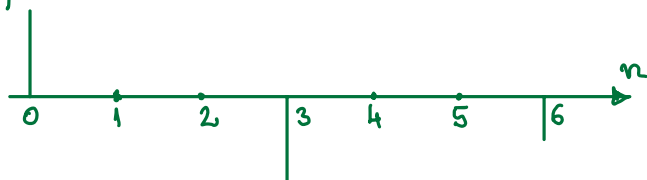
DTFT $\rightarrow$

$X_d(e^{j\omega})$

?

An intermediate step:

$y[n]$



$$y[n] = x[n] \sum_{k=-\infty}^{\infty} \delta[n - kM]$$

Establish: $X(e^{j\omega}) \rightarrow Y(e^{j\omega}) \rightarrow X_d(e^{j\omega})$

Time-domain relationship:

$$y[n] = x[n] \sum_{k=-\infty}^{\infty} \delta[n - kM] = x[n] \frac{1}{M} \sum_{k=0}^{M-1} e^{j\frac{2\pi}{M}kn}$$

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \left(x[n] \frac{1}{M} \sum_{k=0}^{M-1} e^{j\frac{2\pi}{M}kn} \right) e^{-j\omega n}$$

$$= \frac{1}{M} \sum_{k=0}^{M-1} \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega - \frac{2\pi}{M}k)n} = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi}{M}k)})$$

$$\underbrace{\hspace{10em}}_{= X(e^{j(\omega - \frac{2\pi}{M}k)})}$$

$$x_d[n] = y[Mn] \Rightarrow X_d(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\frac{\omega}{M} - \frac{2\pi}{M}k)})$$

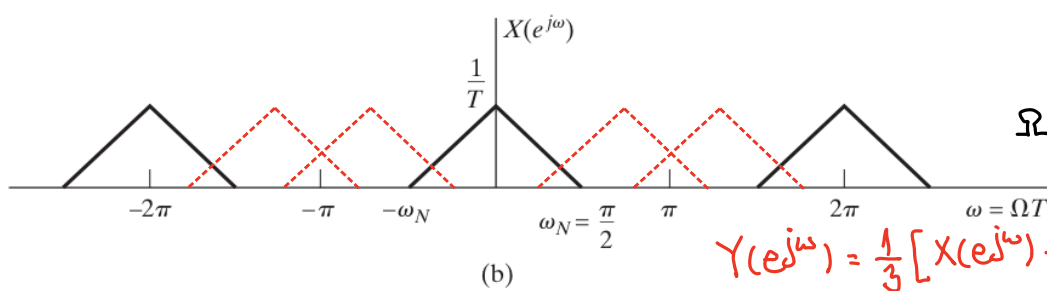
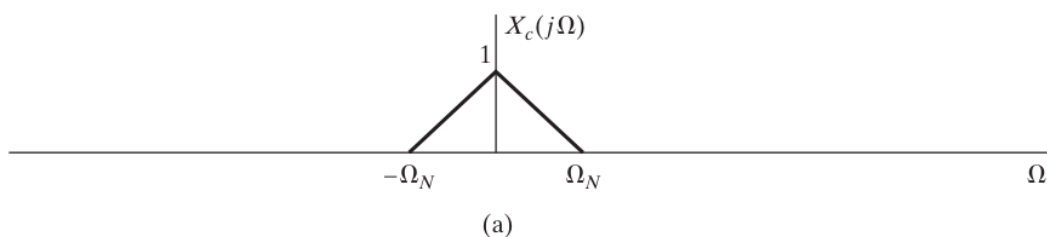
Special case ($M=2$):

$$X_d(e^{j\omega}) = \frac{1}{2} [X(e^{j\frac{\omega}{2}}) + X(e^{j(\frac{\omega}{2}-\pi)})]$$

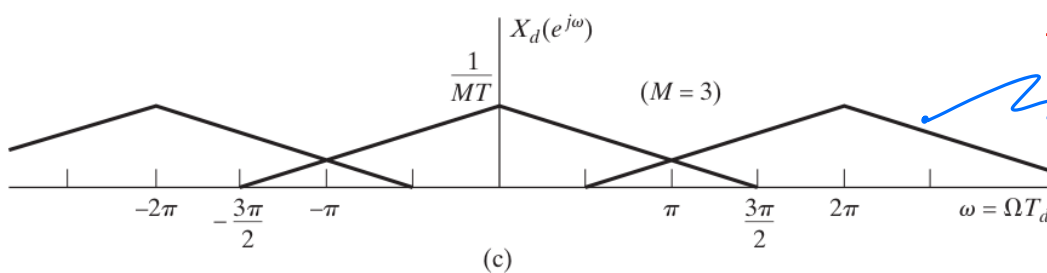
$$X(e^{j\omega}) \rightarrow Y(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi}{M}k)}) \rightarrow$$

$$X(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\frac{\omega}{M} - \frac{2\pi}{M}k)})$$

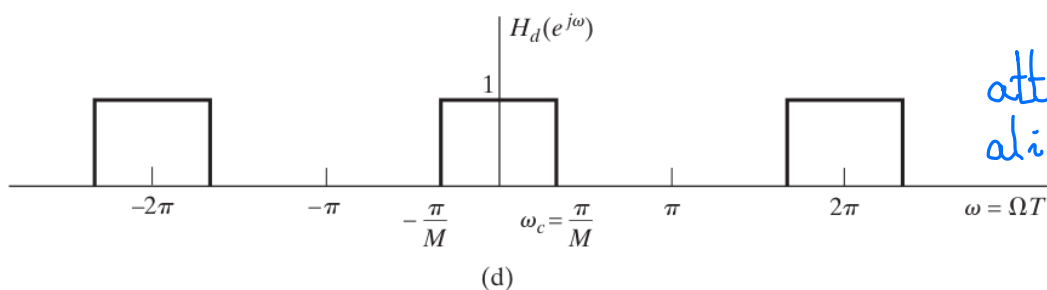
An illustration of downsampling with aliasing:
 $x[n] = x_c(nT)$, downsample with $M=3$



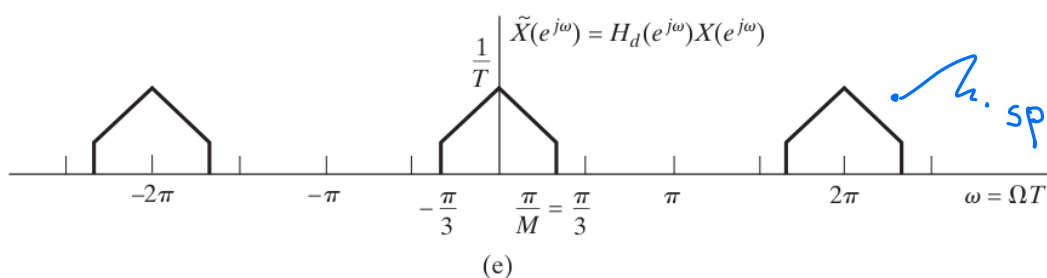
$$Y(e^{j\omega}) = \frac{1}{3} [X(e^{j\omega}) + X(e^{j(\omega - \frac{2\pi}{3})}) + X(e^{j(\omega - \frac{4\pi}{3})})]$$



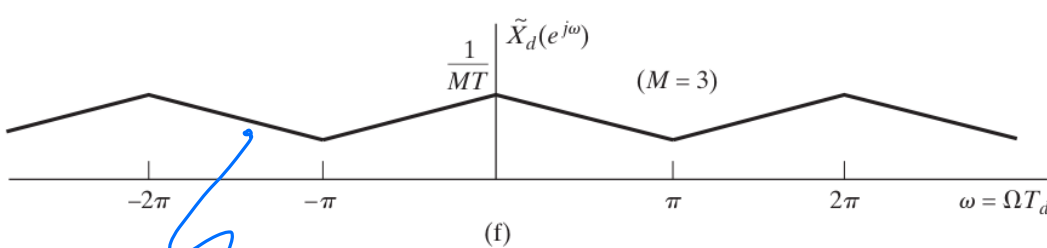
aliasing due to downsampling



attempt to reduce aliasing by filtering $x[n]$ before downsampling

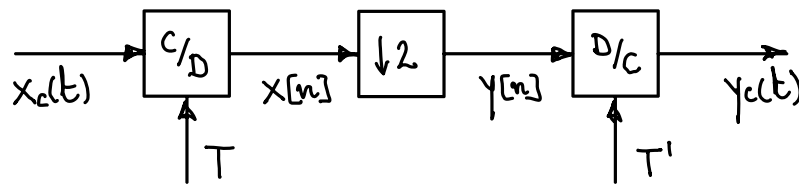


spectrum of $\tilde{x}[n]$

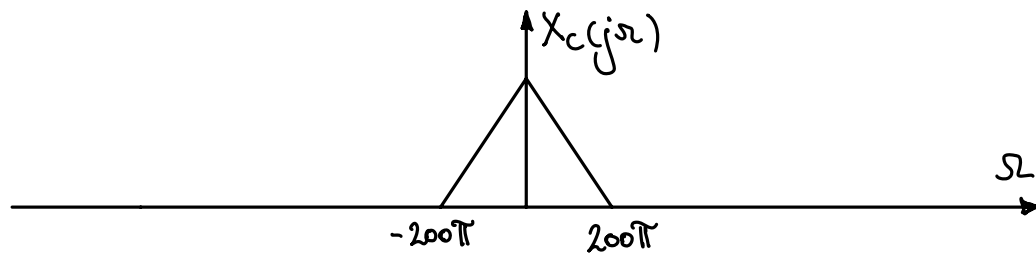


avoid aliasing but affected by distortion

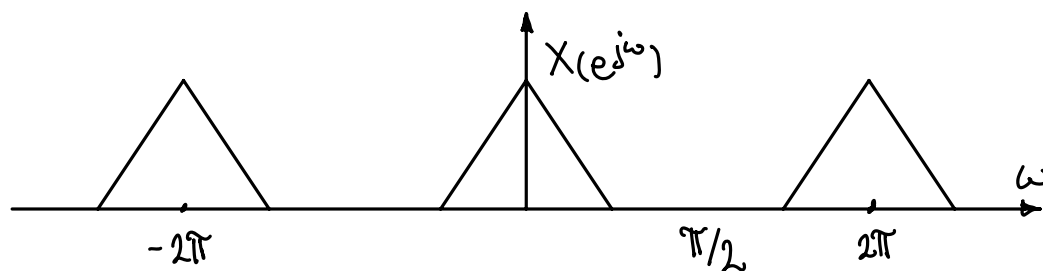
Ex: Consider the system below:



(a) Assume $X_c(j\Omega) = 0$, $|\Omega| > 200\pi$. How should T be chosen so that $X(e^{j\omega}) = 0$ for $\frac{\pi}{2} < |\omega| < \pi$?



$$\Omega_N T < \pi/2$$



$$200\pi \cdot T < \pi/2$$

$$T < 1/400$$

(b) How should T' be chosen so that $y_c(t) = x_c(t)$?

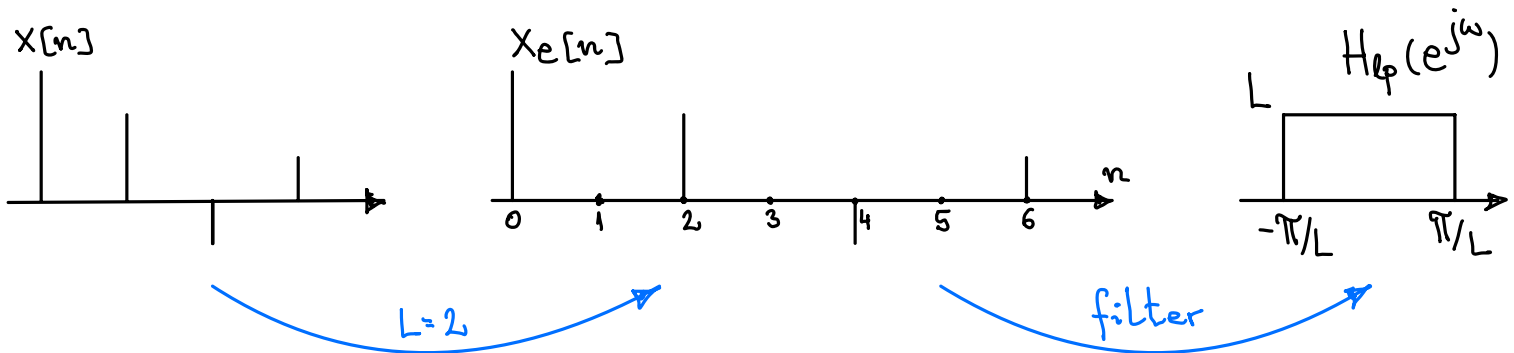
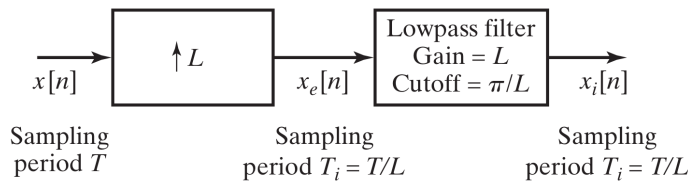
$$M = 2 \quad MT = T' \Rightarrow T' = \frac{1}{200}$$

■ Upsampling: increase the sampling rate by an integer factor

original: $x[n] = x_c(nT)$ $\longrightarrow$ sampled at T

upsampled: $x_u[n] = x_c(nT/L)$ $\longrightarrow$ sampled at T/L

The operation of upsampling:



$$X_e(e^{j\omega}) = ?$$

$$x_e[n] = \begin{cases} x\left[\frac{n}{L}\right], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$$

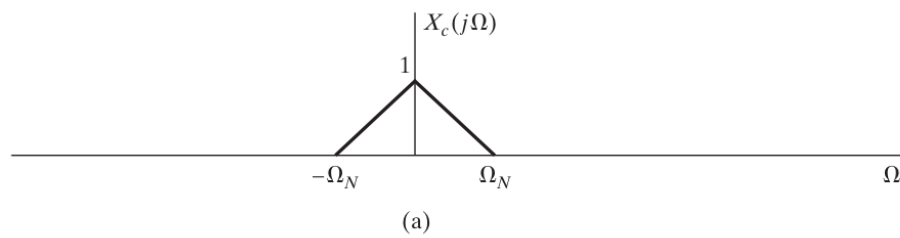
$$x_e[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - kL]$$

$$X_e(e^{j\omega}) = \text{DTFT} \{x_e[n]\} = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k] \delta[n - kL] \right) e^{-j\omega n}$$

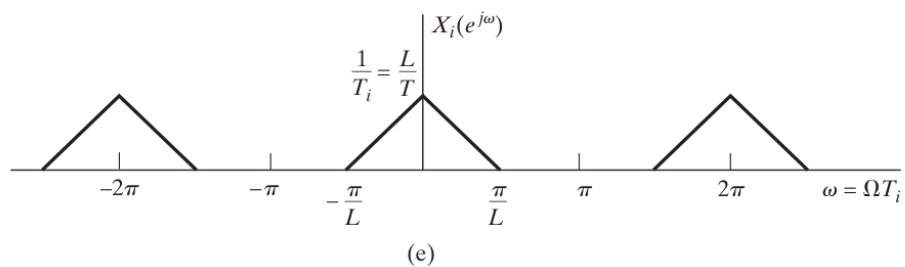
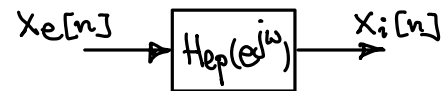
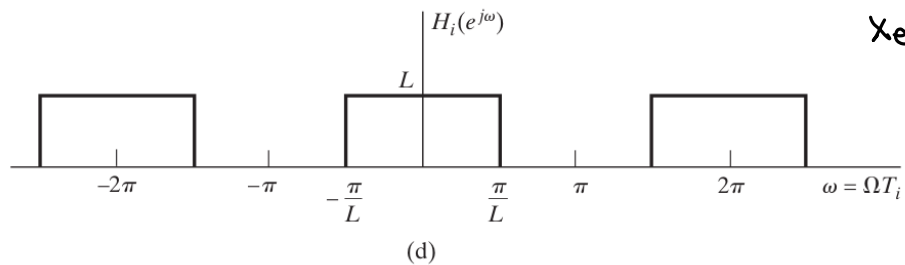
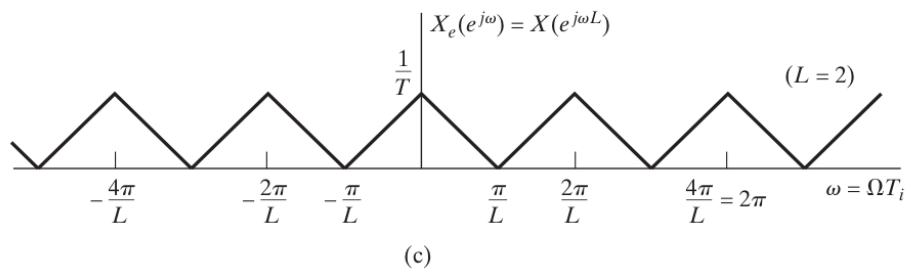
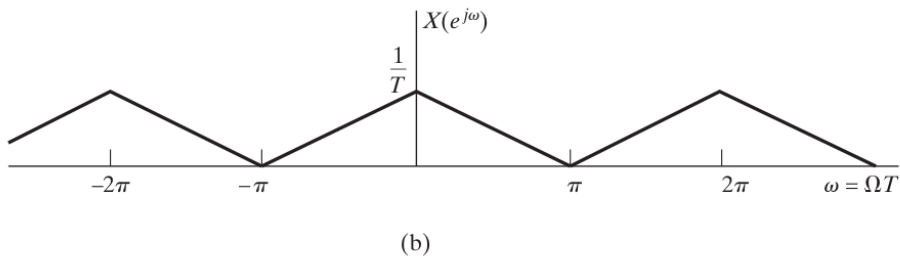
$$= \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} e^{-j\omega n} \delta[n - kL] = \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega kL}$$

$$\Rightarrow X_e(e^{j\omega}) = X(e^{j\omega L})$$

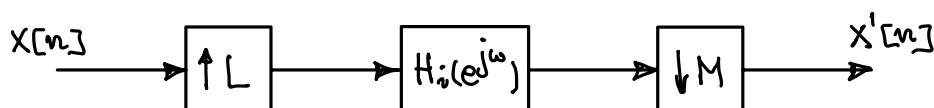
An illustration of upsampling:



$$\Omega_S = 2\Omega_N$$

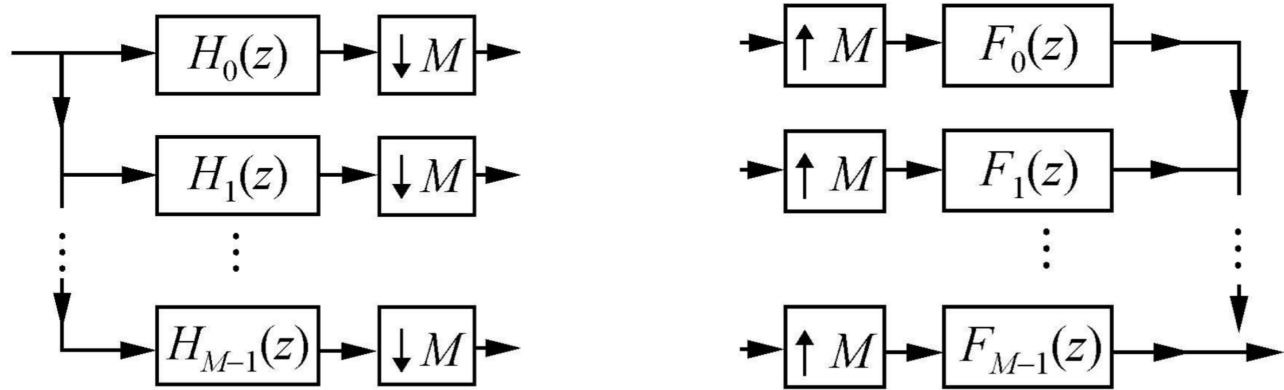


■ Changing the sampling rate by a rational factor



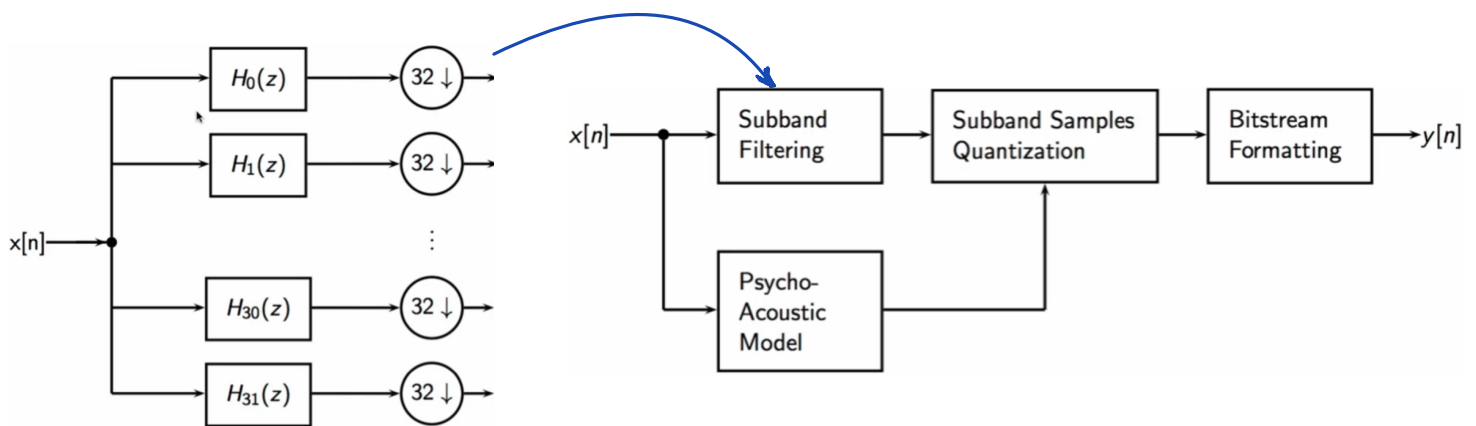
$\frac{M}{L}$: a rational factor ($\frac{M}{L} > 1 \Rightarrow$ reduction of sampling rate)

A digression: filter banks, broadly used tool



Applications: graphic EQ, signal compression, features extraction for machine learning tasks (e.g. genre classification)

Ex: MP3 compression



$H_0(e^{j\omega})$:

